Rehmat Ali

(Embedded AI Engineer)

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Summary

Master's student at Polytech Nantes specializing in AI, Embedded Systems, and Edge AI, with 3.5 years of software development experience in ERP systems. Currently applying ML model XGBoost to predict FPGA circuit-generation parameters for complex routing congestion benchmarks.

Education

Master's In Embedded Technologies and Artificial Intelligence Nantes Université, Nantes, France — Nov. 2025 to Jul. 2026

Majors: AI for Advanced Digital Communication, AI Fundamentals, Architecture of Embedded Systems, Embedded OS, Edge Computing, Design of Hardware/Software Architecture, Advanced Signal Processing, Antenna Processing, Power Consumption and Reliability, Design of Embedded Systems and Embedded Software.

Bachelor of Science (Software Engineering) SZABIST, Islamabad — Sept. 2017 to July 2021

GPA: 3.16/4.0
Majors: Data Science, Digital Image Processing, Artificial Intelligence, Software Engineering, Object Oriented Programming, Software Architecture & Design and Visual Programming,

Internship

Institut d'Électronique et des Technologies du numéRique (IETR), FPGA Benchmarking Framework for AI-Based EDA Tools — Nantes, France — Feb. 2026 to Jul. 2026 — 6 Months

- Built an AI/ML-based FPGA benchmarking framework using XGBoost.
- Created ML-ready Vivado datasets from congestion, routing, utilization, and runtime metrics.
- Developed an AI-assisted adversarial netlist generation flow for EDA benchmarking.
- Designed a Python/NetworkX graph generator for FPGA-aware circuit generation.
- Automated Verilog/XDC generation for AI-driven FPGA routing-stress testing.

Experience

ERPHUB, ERP Solution Devleoper — Cleveland, USA — Onsite: Islamabad — Jan 2023 to Aug 2025 — 2.8 years

- Implemented AI-driven Zoho solutions, including predictive lead scoring, chatbots, smart automation, and business forecasting.
- Developed custom Zoho applications and automated business workflows using Zoho Creator and Deluge.
- Optimized processes across Zoho CRM, Books, People, Desk, and Analytics.
- Integrated Zoho applications with external platforms using REST APIs, webhooks, and custom functions.

Link: Experience Letter

FairchanceforCRM Zoho Developer — Lahore, Pakistan

Onsite

Aug 2021 to Dec 2022

1.5 years

- Held regular meetings with clients to gather and analyze business requirements for Zoho-based solutions.
- Planned and implemented customized workflows across various Zoho applications to streamline business processes.
- Designed and developed custom workflows and functions to automate repetitive tasks and enhance CRM efficiency.
- Developed and trained a SalesIQ chatbot to handle multiple customer scenarios, trigger actions, and provide contextual responses, improving the overall digital support experience.

Link: Experience Letter

Projects

Deep Learning Based Framework for Colorectal Polyp Classification *September 2021*
Type: Academia
Members: 2
Role: Team Lead
Description: Built a CNN-based web system for colorectal polyp classification, achieving 94% accuracy and generating diagnostic reports from colonoscopy images.
Link: Documentation

Voice Controlled Home Automation using Arduino *December 2020*
Type: Academia
Members: Individual Project
Role: Developer
Description: Developed an Arduino-based home automation system to control electronic devices using voice commands through a Bluetooth-connected mobile application. Implemented C++ logic for command processing and device control.

GiftTrees Full-Stack Business Management System *February 2024 – July 2026*
Type: Professional Project
Role: Developer
Description: Developed and configured the complete frontend and backend system for GiftTrees, including customer management, admin dashboards, reporting modules, commission calculation, and custom control applications for business operations.
Website: gifttrees.com

Technologies

Programming: Verilog, Python, PHP, SQL, C/C++, JavaScript, Deluge
FPGA & Embedded: Vivado, XDC, routing congestion analysis, Arduino, edge computing
AI/ML: XGBoost, CNNs, TensorFlow, model training, dataset preparation, evaluation
Web/ERP: HTML, CSS, Bootstrap, jQuery, Zoho Creator/CRM/Analytics, REST APIs, webhooks

Languages

English (Fluent) Urdu (Native) French (Basic)

REFERENCES

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